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ELECTRIC CONNECTOR AND ELECTRIC CONNECTOR ASSEMBLY WITH SIMPLIFIED GROUNDING METHOD AND IMPROVED STABILITY OF SIGNAL TRANSMISSION

Abstract

An electrical connector includes an insulating body, a number of first conductive terminals, a number of second conductive terminals, and a number of cables. The first conductive terminals include a number of first signal terminals and a number of first ground terminals. The second conductive terminals include a number of second signal terminals and a number of second ground terminals. The cables include a number of first cables electrically connected with the first signal terminals and a number of second cables electrically connected with the second signal terminals. The first ground terminals and the second ground terminals are connected to each other and are adapted to be electrically connected to a circuit board. With this arrangement, the present disclosure avoids ground return through cables, thereby simplifying the grounding method and improving the stability of signal transmission. An electrical connector assembly having the electrical connector is also disclosed.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 17/890,051, filed on Aug. 17, 2022, which claims priority of a Chinese Patent Application No. 202122085190.X, filed on Aug. 31, 2021 and titled “ELECTRIC CONNECTOR AND ELECTRIC CONNECTOR ASSEMBLY”, the entire content of which is incorporated herein by reference.

TECHNICAL FIELD
[0002] The present disclosure relates to an electrical connector and an electrical connector assembly, which belongs to a technical field of electrical connectors.
BACKGROUND
[0003] An electrical connector usually includes an insulating body and a plurality of conductive terminals fixed to the insulating body. The conductive terminal includes a mating portion for contacting with a mating connector. The plurality of conductive terminals include high-speed signal terminals, ground terminals and other terminals. The high-speed signal terminals and the ground terminals are adapted for connecting with cables, and the other terminals are adapted for being mounted to a circuit board.
[0004] With this design, the ground terminals need to be grounded back through the cables or used as a reference, and the length of the cable will be relatively long, which affects the stability of signal transmission.
SUMMARY
[0005] The present disclosure adopts the following technical solution: an electrical connector adapted to be at least partially mounted to a circuit board, the electrical connector including: an insulating body, the insulating body defining a mating slot; a plurality of first conductive terminals, the plurality of first conductive terminals including a plurality of first signal terminals and a plurality of first ground terminals, each of the first conductive terminals including a first mating portion extending into the mating slot, the first mating portions of the plurality of first conductive

terminals being arranged in a first row; a plurality of second conductive terminals, the plurality of second conductive terminals including a plurality of second signal terminals and a plurality of second ground terminals, each of the second conductive terminals including a second mating portion extending into the mating slot, the second mating portions of the plurality of second conductive terminals being arranged in a second row; and a plurality of cables, the plurality of cables including a plurality of first cables electrically connected with the plurality of first signal terminals and a plurality of second cables electrically connected with the plurality of second signal terminals; wherein the plurality of first ground terminals and the plurality of second ground terminals are connected to each other, and are adapted to be electrically connected to the circuit board; wherein the plurality of first conductive terminals further include a plurality of first intermediate terminals, each of the first intermediate terminals includes a first mounting foot for being mounted to the circuit board; wherein the plurality of second conductive terminals further include a plurality of second intermediate terminals, each of the second intermediate terminals includes a second mounting foot for being mounted to the circuit board; wherein in a width direction of the insulating body, the plurality of first intermediate terminals are aligned with the plurality of first ground terminals, and the plurality of second intermediate terminals are aligned with the plurality of second ground terminals; wherein the electrical connector further includes a first tail portion; the first tail portion is connected to at least one of the plurality of first ground terminals; and wherein the electrical connector further includes a second tail portion; the second tail portion is connected to at least one of the plurality of second ground terminals.
[0006] The present disclosure adopts the following technical solution: an electrical connector assembly, including: a first electrical connector; a second electrical connector; and a circuit board, the circuit board including a first side and a second side opposite to the first side; wherein the first electrical connector is mounted to the first side of the circuit board, and the second electrical connector is mounted to the second side of the circuit board; wherein the first electrical connector includes: an insulating body, the insulating body defining a mating slot; a plurality of first conductive terminals, the plurality of first conductive terminals including a plurality of first signal terminals and a plurality of first ground terminals, each of the first conductive terminals including a first mating portion extending into the mating slot, the first mating portions of the plurality of first conductive terminals being arranged in a first row; a plurality of second conductive terminals, the plurality of second conductive terminals including a plurality of second signal terminals and a plurality of second ground terminals, each of the second conductive terminals including a second mating portion extending into the mating slot, the second mating portions of the plurality of second conductive terminals being arranged in a second row; and a plurality of cables, the plurality of cables including a plurality of first cables electrically connected with the plurality of first signal terminals and a plurality of second cables electrically connected with the plurality of second signal terminals; wherein the plurality of first ground terminals and the plurality of second ground terminals are connected to each other and are electrically connected to the circuit board; wherein the first electrical connector further includes a first tail portion integrally connected with at least two of the plurality of first ground terminals and a second tail portion integrally connected with at least two of the plurality of second ground terminals; wherein the first tail portion is in contact with the second tail portion; and the first electrical connector further includes a ground mounting portion integrally extending from one of the first tail portion and the second tail portion, and the ground mounting portion is mounted to the circuit board; a remaining one of the first tail portion and the second tail portion does not provide any ground mounting portion mounted to the circuit board; and the remaining one of the first tail portion and the second tail portion is electrically connected to the circuit board via the ground mounting portion provided on the one of the first tail portion and the second tail portion.
[0007] The present disclosure adopts the following technical solution: an electrical connector configured to be at least partially mounted to a circuit board, the electrical connector including: an

insulating body, the insulating body defining a mating slot; a plurality of first conductive terminals, the plurality of first conductive terminals including a plurality of first signal terminals and a plurality of first ground terminals, each of the first conductive terminals including a first mating portion extending into the mating slot, the first mating portions of the plurality of first conductive terminals being arranged in a first row; a plurality of second conductive terminals, the plurality of second conductive terminals including a plurality of second signal terminals and a plurality of second ground terminals, each of the second conductive terminals including a second mating portion extending into the mating slot, the second mating portions of the plurality of second conductive terminals being arranged in a second row; a plurality of cables, the plurality of cables including a plurality of first cables connected to the plurality of first signal terminals and a plurality of second cables connected to the plurality of second signal terminals; a first tail portion connecting at least two of the plurality of first ground terminals; a second tail portion connecting at least two of the plurality of second ground terminals, the first tail portion being in contact with the second tail portion; and a ground mounting portion extending from one of the first tail portion and the second tail portion; wherein the ground mounting portion is configured to be mounted to the circuit board; a remaining one of the first tail portion and the second tail portion does not provide any ground mounting portion configured to be mounted to the circuit board; and the remaining one of the first tail portion and the second tail portion is electrically connected to the circuit board via the ground mounting portion provided on the one of the first tail portion and the second tail portion.

Description

BRIEF DESCRIPTION OF DRAWINGS
[0008] FIG. **1** is a perspective schematic view of an electrical connector assembly in accordance with an embodiment of the present disclosure;
[0009] FIG. **2** is a right side view of FIG. **1**;
[0010] FIG. **3** is a partially exploded perspective view of FIG. **1**;
[0011] FIG. **4** is a partially exploded perspective view of FIG. **3** from another angle;
[0012] FIG. **5** is a partial perspective exploded view of an electrical connector in FIG. **3**;
[0013] FIG. **6** is a partially exploded perspective view of FIG. **5** from another angle;
[0014] FIG. **7** is a further perspective exploded view after removing a metal cage in FIG. **5**;
[0015] FIG. **8** is a further perspective exploded view of FIG. **7**;
[0016] FIG. **9** is a partially exploded perspective view of FIG. **8** from another angle;
[0017] FIG. **10** is a further perspective exploded view after removing an insulating body and an over-molding housing in FIG. **8**; FIG.
[0018] FIG. **11** is a perspective exploded view of FIG. **10** from another angle;
[0019] FIG. **12** is a perspective view of a plurality of first conductive terminals and a plurality of second conductive terminals;
[0020] FIG. **13** is a perspective schematic view of FIG. **12** from another angle;
[0021] FIG. **14** is a perspective schematic view of a mutual positional relationship of the first conductive terminals, a first ground piece, first cables, the second conductive terminals, a second ground piece, and second cables when they are assembled in position;
[0022] FIG. **15** is a perspective schematic view of FIG. **14** from another angle;
[0023] FIG. **16** is a right side view of FIG. **14**;
[0024] FIG. **17** is a partial exploded view of FIG. **16**;
[0025] FIG. **18** is a further exploded view of FIG. **17**;
[0026] FIG. **19** is a top view of FIG. **17**;
[0027] FIG. **20** is a perspective exploded view of FIG. **14**;
[0028] FIG. **21** is a schematic cross-sectional view taken along line A-A in FIG. **1**;

[0029] FIG. **22** is a schematic cross-sectional view taken along line B-B in FIG. **1**;
[0030] FIG. **23** is a schematic cross-sectional view taken along the line C-C in FIG. **1**; and
[0031] FIG. **24** is a perspective view of the first conductive terminals, the second conductive terminals, a first fixing block, a second fixing block and a fixing piece, wherein the first fixing block and the second fixing block are separated from the first conductive terminals and the second conductive terminals, respectively.

DETAILED DESCRIPTION

[0032] Exemplary embodiments will be described in detail here, examples of which are shown in drawings. When referring to the drawings below, unless otherwise indicated, same numerals in different drawings represent the same or similar elements. The examples described in the following exemplary embodiments do not represent all embodiments consistent with this application. Rather, they are merely examples of devices and methods consistent with some aspects of the application as detailed in the appended claims.

[0033] The terminology used in this application is only for the purpose of describing particular embodiments, and is not intended to limit this application. The singular forms “a”, “said”, and “the” used in this application and the appended claims are also intended to include plural forms unless the context clearly indicates other meanings.

[0034] It should be understood that the terms “first”, “second” and similar words used in the specification and claims of this application do not represent any order, quantity or importance, but are only used to distinguish different components. Similarly, “an” or “a” and other similar words do not mean a quantity limit, but mean that there is at least one; “multiple” or “a plurality of” means two or more than two. Unless otherwise noted, “front”, “rear”, “lower” and/or “upper” and similar words are for ease of description only and are not limited to one location or one spatial orientation. Similar words such as “include” or “comprise” mean that elements or objects appear before “include” or “comprise” cover elements or objects listed after “include” or “comprise” and their equivalents, and do not exclude other elements or objects. The term “a plurality of” mentioned in the present disclosure includes two or more.

[0035] Hereinafter, some embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. In the case of no conflict, the following embodiments and features in the embodiments can be combined with each other.

[0036] Referring to FIGS. **1** to **4**, the present disclosure discloses an electrical connector assembly **400** including a circuit board **300**, a first electrical connector **100**, and a second electrical connector **200**. The circuit board **300** includes a first side **301** (for example, an upper surface) and a second side **302** (for example, a lower surface) opposite to the first side **301**. The circuit board **300** also defines a plurality of locking holes **303** extending through the first side **301** and the second side **302**, and a plurality of conductive holes **304** extending through the first side **301** and the second side **302**. In the illustrated embodiment of the present disclosure, the first electrical connector **100** is mounted to the first side **301** of the circuit board **300**. The second electrical connector **200** is mounted to the second side **302** of the circuit board **300**. This arrangement is beneficial to improve the utilization rate of the circuit board **300**. Of course, in other embodiments, the electrical connector assembly **400** may also include the circuit board **300** and one connector (for example, the first electrical connector **100**).

[0037] Referring to FIGS. **5** to **11**, the first electrical connector **100** and/or the second electrical connector **200** are high-speed connectors, such as QSFP connectors, SFP connectors, OSFP connectors, or OSFP-DD connectors, etc. Of course, in other embodiments, the first electrical connector **100** and/or the second electrical connector **200** may also be other types of connectors. A general concept of the first electrical connector **100** and the second electrical connector **200** is an electrical connector. In the illustrated embodiment of the present disclosure, the first electrical connector **100** and the second electrical connector **200** have the same structure, therefore the following only takes the first electrical connector **100** as an example for detail portioned description.

[0038] The first electrical connector **100** includes an insulating body **3**, a first conductive terminal module **1** mounted to the insulating body **3**, a second conductive terminal module **2** mounted to the insulating body **3**, a fixing piece **4** fixed on the first conductive terminal module **1** and the second conductive terminal module **2**, an over-molding housing **5** at least partially over-molded on the insulating body **3**, and a metal cage **6** enclosing the insulating body **3** and the over-molding housing **5**.

[0039] The insulating body **3** includes a base **31** and a protruding portion **32** extending forwardly from the base **31**. The insulating body **3** includes a mounting wall **331** at the bottom. The mounting wall **331** includes a mounting surface **3311** for being mounted to the circuit board **300** and a plurality of positioning posts **3312** protruding beyond the mounting surface **3311** for being mounted to the circuit board **300**. The mounting wall **331** further includes a mounting groove **3313** and a plurality of positioning protrusions **3314** protruding into the mounting groove **3313**. The protruding portion **32** includes a mating surface **321** and a mating slot **320** extending through the mating surface **321**. The base **31** includes a rear end surface **311** opposite to the mating surface **321** and a receiving space **310** extending through the rear end surface **311**. The receiving space **310** is in communication with the mating slot **320** and is adapted for receiving the first conductive terminal module **1** and the second conductive terminal module **2**.

[0040] The first conductive terminal module **1** includes a first insulating block **11**, a plurality of first conductive terminals **12** fixed to the first insulating block **11**, a plurality of first cables **13**, a first fixing block **14** for fixing at least parts of the first conductive terminals **12**, and a first ground piece **15**.

[0041] The first insulating block **11** includes a first slot **111** and a first mating surface **112**. In an embodiment of the present disclosure, the plurality of first conductive terminals **12** are insert-molded in the first insulating block **11**. Parts of the plurality of first conductive terminals **12** are exposed in the first slot **111**. Referring to FIGS. **12** to **23**, the plurality of first conductive terminals **12** include a plurality of first signal terminals S**1**, a plurality of first ground terminals G**1**, and a plurality of first intermediate terminals M**1**. In an embodiment of the present disclosure, the plurality of first signal terminals S**1** include a plurality of first signal terminal groups. Each first signal terminal group includes two of the first signal terminals S**1**. Each first signal terminal group is located between two first ground terminals G**1**. The first signal terminals S**1** and the first cables **13** are fixed by soldering or welding.

[0042] Specifically, each first conductive terminal **12** includes a first mating portion **121** extending into the mating slot **320**. The first mating portions **121** of the first conductive terminals **12** are arranged in a first row (for example, an upper row). The first conductive terminal module **1** includes a first tail portion **122** integrally connected with the plurality of first ground terminals G**1**. In the illustrated embodiment of the present disclosure, the first tail portion **122** is bent and extends toward the first mating surface **112**. Referring to FIG. **13**, in the illustrated embodiment of the present disclosure, the plurality of first ground terminals G**1** are two groups. Correspondingly, there are two first tail portions **122**. Each first tail portion **122** extends along a width direction W-W of the insulating body **3**. The plurality of first ground terminals G**1** of each group are integrally connected to the corresponding first tail portion **122** at intervals.

[0043] The first ground piece **15** is mounted in the first slot **111**. The first ground piece **15** is in contact with the plurality of first ground terminals G**1** so as to connect the plurality of first ground terminals G**1** in series, thereby increasing the grounding area and improving the grounding effect. The first ground piece **15** does not contact the first signal terminals S**1** in order to avoid short circuits. Specifically, the first ground piece **15** includes a plurality of first mating portions **151** in contact with the corresponding first ground terminals G**1** and a plurality of first protrusions **152** each of which is connected between two adjacent first matting portions **151**. A cross-section of each first protrusion **152** is substantially U-shaped. The first protrusion **152** defines a first space **1520**.

The first signal terminal S**1** is located in the corresponding first space **1520** to avoid contact with any part of the first ground piece **15**.

[0044] Each of the first intermediate terminals M**1** includes a first extension portion **123** extending beyond the insulating body **3** and a first bending portion **124** vertically bent from a rear end of the first extension portion **123**. The first bending portion **124** includes a first body portion **1241**, a first deflection portion **1242** bent from the first body portion **1241**, and a first end **1243** extending from the first deflection portion **1242**. The first end **1243** is parallel to the first main body **1241**. The first deflection portion **1242** is perpendicular to the first end **1243** and the first body portion **1241**. The first end **1243** includes a first mounting foot **1244** for mounting the first intermediate terminal M**1** to the circuit board **300**. The first mounting foot **1244** includes a through hole **1245**, so that the first mounting foot **1244** has a deformability to be better engaged with the corresponding conductive hole **304** of the circuit board **300**.

[0045] In order to facilitate the arrangement, deflection directions of the first deflection portions **1242** of any two adjacent first intermediate terminals M**1** are opposite, so that the first mounting feet **1244** of any two adjacent first intermediate terminals M**1** are alternately arranged in two rows.

[0046] Similarly, the second conductive terminal module **2** includes a second insulating block **21**, a plurality of second conductive terminals **22** fixed to the second insulating block **21**, a plurality of second cables **23**, a second fixing block **24** for fixing at least parts of the second conductive terminals **22**, and a second ground piece **25**.

[0047] The second insulating block **21** includes a second slot **211** and a second mating surface **212**. In an embodiment of the present disclosure, the plurality of second conductive terminals **22** are insert-molded in the second insulating block **21**. Parts of the plurality of second conductive terminals **22** are exposed in the second slot **211**. Referring to FIGS. **12** to **23**, the plurality of second conductive terminals **22** include a plurality of second signal terminals S**2**, a plurality of second ground terminals G**2**, and a plurality of second intermediate terminals M**2**. In an embodiment of the present disclosure, the plurality of second signal terminals S**2** include a plurality of second signal terminal groups. Each second signal terminal group includes two of the second signal terminals S**2**. Each second signal terminal group is located between the two second ground terminals G**2**. The second signal terminals S**2** and the second cables **23** are fixed by soldering or welding. In some embodiments, the plurality of first signal terminals S**1** and the plurality of second signal terminals S**2** are adapted to transmit relatively high-speed signals. The plurality of first intermediate terminals M**1** and the plurality of second intermediate terminals M**2** are adapted to transmit relatively non-high-speed signals.

[0048] Each second conductive terminal **22** includes a second mating portion **221** extending into the mating slot **320**. The second mating portions **221** of the second conductive terminals **22** are arranged in a second row (for example, a lower row). The first mating portions **121** located in the first row and the second mating portions **221** located in the second row are parallel to each other so as to jointly clamp a tongue plate (not shown) of a mating connector. In some embodiments, the first mating portions **121** and the second mating portions **221** have elasticity, which can better clamp the tongue plate of the mating connector and elastically abut against conductive pads of the tongue plate. The second conductive terminal module **2** includes a second tail portion **222** integrally connected with the plurality of second ground terminals G**2**. In the illustrated embodiment of the present disclosure, the second tail portion **222** is bent and extends toward the second mating surface **212**. Referring to FIG. **13**, in the illustrated embodiment of the present disclosure, the plurality of second ground terminals G**2** are two groups. Correspondingly, there are two second tail portions **222**. Each second tail portion **222** extends along the width direction W-W of the insulating body **3**. The plurality of second ground terminals G**2** of each group are integrally connected to the corresponding second tail portion **222** at intervals.

[0049] The second ground piece **25** is mounted in the second slot **211**, and the second ground piece **25** is in contact with the plurality of second ground terminals G**2** to connect the plurality of second ground terminals G**2** in series, thereby increasing the grounding area and improving the grounding effect. The second ground piece **25** does not contact the plurality of second signal terminals S**2** in order to avoid short circuits. Specifically, the second ground piece **25** includes a plurality of second mating portions **251** in contact with the corresponding second ground terminals G**2** and a plurality of second protrusions **252** each of which is connected between two adjacent second mating portions **251**. A cross-section of the second protrusion **252** is substantially U-shaped. The second protrusion **252** defines a second space **2520**. The second signal terminal S**2** is located in the corresponding second space **2520** to avoid contact with any part of the second ground piece **25**.

[0050] Each of the second intermediate terminals M**2** includes a second extension portion **223** extending beyond the insulating body **3** and a second bending portion **224** vertically bent from a rear end of the second extension portion **223**. The second bending portion **224** includes a second body portion **2241**, a second deflection portion **2242** bent from the second body portion **2241**, and a second end **2243** extending from the second deflection portion **2242**. The second end **2243** is parallel to the second body portion **2241**. The second deflection portion **2242** is perpendicular to the second end **2243** and the second body portion **2241**. The second end **2243** includes a second mounting foot **2244** for mounting the second middle terminal M**2** to the circuit board **300**. The second mounting foot **2244** includes a through hole **2245**, so that the second mounting foot **2244** has a deformability to be better engaged with the corresponding conductive hole **304** of the circuit board **300**.

[0051] In order to facilitate the arrangement, deflection directions of the second deflection portions **2242** of any two adjacent second intermediate terminals M**2** are opposite, so that the second mounting feet **2244** of any two adjacent second intermediate terminals M**2** are alternately arranged in two rows.

[0052] The first electrical connector **100** further includes a ground mounting portion **16** integrally extending from the first tail portion **122** and/or the second tail portion **222**. The ground mounting portion **16** is adapted for being mounted to the circuit board **300**. Referring to FIG. **24**, in the illustrated embodiment of the present disclosure, the ground mounting portion **16** includes a first ground foot **1601** and a second ground foot **1602**. The first ground foot **1601** is integrally connected to one of the first tail portions **122**. The second ground foot **1602** is integrally connected to one of the second tail portions **222**. One group of the plurality of first ground terminals G**1** and one group of the plurality of second ground terminals G**2** are jointly and are adapted to be electrically connected to the circuit board **300** via the first ground foot **1601**. The plurality of first ground terminals G**1** in the other group and the plurality of second ground terminals G**2** in the other group are jointly and are adapted to be electrically connected to the circuit board **300** via the second ground foot **1602**. The ground mounting portion **16** includes a through hole **161**, so that the ground mounting portion **16** has a deformability to be better engaged with the corresponding conductive hole **304** of the circuit board **300**.

[0053] In order to facilitate the arrangement, the first mounting feet **1244** and the first ground feet **1601** of any two adjacent first intermediate terminals M**1** are alternately arranged in two rows. The second mounting feet **2244** and the second ground feet **1602** of any two adjacent second middle terminals M**2** are alternately arranged in two rows. In the illustrated embodiment of the present disclosure, the plurality of first bending portions **124** and the first ground foot **1601** are fixed to the first fixing block **14**, and the plurality of second bending portions **224** and the second ground foot **1602** are fixed to the second fixing block **24** in order to facilitate the improvement of the structural strength of the terminals and the maintenance of the distance between the terminals.

[0054] In the width direction W-W, the plurality of first intermediate terminals M**1** are located between the two groups of the plurality of first ground terminals G**1**, and the plurality of second intermediate terminals M**2** are located between the two groups of the plurality of second ground terminals G**2**. The two groups of the plurality of first ground terminals G**1** are respectively integrally connected to the two first tail portions **122**. The two groups of the plurality of second

ground terminals G**2** are respectively integrally connected to the two second tail portions **222**. The two first tail portions **122** are in contact with the two second tail portions **222**, respectively.
[0055] In the illustrated embodiment of the present disclosure, after the first conductive terminal module **1** and the second conductive terminal module **2** are assembled in the receiving space **310** of the insulating body **3**, the first mating surface **112** of the first insulating block **11** and the second mating surface **212** of the second insulating block **21** are adjacent to each other. At this time, the first tail portion **122** of the first conductive terminal module **1** is in contact with the second tail portion **222** of the second conductive terminal module **2** so as to increase the grounding area and improve the grounding effect. With the help of the first ground piece **15** and the second ground piece **25**, the first ground terminals G**1** of the first conductive terminal module **1** and the second ground terminals G**2** of the second conductive terminal module **2** are connected in series. In addition, the first ground terminals G**1**, the first ground piece **15**, the second ground terminals G**2**, and the second ground piece **25** can all be grounded to the circuit board **300** through the ground mounting portion **16**. Compared with the related art, the first ground terminals G**1** and the second ground terminals G**2** are not grounded through cables, which avoids using cables for ground return. On one hand, it simplifies the grounding method; on the other hand, it is beneficial to improve the stability of signal transmission.
[0056] In the illustrated embodiment of the present disclosure, the first mounting feet **1244** of the first intermediate terminals M**1** are alternately arranged in two rows. The second mounting feet **2244** of the second middle terminals M**2** are alternately arranged in two rows. The first mounting feet **1244** are located at a rear end of the second mounting feet **2244** along a mating direction of the mating connector. In other words, the first mounting feet **1244** of the first intermediate terminals M**1** and the second mounting feet **2244** of the second intermediate terminals M**2** are arranged in four rows. The first mounting feet **1244** are located at the rear end of the second mounting feet **2244** along the mating direction of the mating connector.
[0057] Referring to FIG. **9**, the fixing piece **4** includes a first positioning portion **41** received in the mounting groove **3313** and a second positioning portion **42** connected to the first positioning portion **41**. The first positioning portion **41** includes a plurality of positioning holes **411** matched with the positioning protrusions **3314**. The second positioning portion **42** includes a first positioning slot **421** sleeved on the first fixing block **14** and a second positioning slot **422** sleeved on the second fixing block **24**. The first positioning slot **421** and the second positioning slot **422** are closely matched with the first fixing block **14** and the second fixing block **24**, respectively, so as to limit the first fixing block **14** and the second fixing block **24**. As a result, a distance between the first fixing block **14** and the second fixing block **24** is restricted, thereby maintaining a distance between the first conductive terminals **12** and the second conductive terminals **22**.
[0058] Referring to FIGS. **5** and **6**, the metal cage **6** includes a top wall **61**, a bottom wall **62**, a first side wall **63** connected to one side of the top wall **61** and one side of the bottom wall **62**, a second side wall **64** connected to the other side of the top wall **61** and the other side of the bottom wall **62**, and a rear wall **65** at a rear end. The metal cage **6** is roughly cylindrical, so as to have a better shielding effect. The top wall **61**, the bottom wall **62**, the first side wall **63** and the second side wall **64** are jointly enclosed to form a receiving cavity **60** for receiving most part of the mating connector. Referring to FIG. **23**, the receiving cavity **60** is in communication with the mating slot **320**. The receiving cavity **60** is located at a front end of the mating slot **320**. The receiving cavity **60** and the mating slot **320** are adapted for jointly receiving the mating connector.
[0059] Referring to FIGS. **3** to **6**, the first side wall **63** of the metal cage **6** is provided with a first side edge **631**, a plurality of first locking feet **632** protruding from the first side edge **631**, and a plurality of first buckling feet **633** protruding from the first side edge **631**. The first locking feet **632** and the first bucking feet **633** are adapted for being inserted into the corresponding locking holes **303** of the circuit board **300** and being fixed to the circuit board **300**.
[0060] The second side wall **64** of the metal cage **6** is provided with a second side edge **641**, a

plurality of second locking feet **642** protruding from the second side edge **641**, and a plurality of second buckling feet **643** protruding from the second side edge **641**. Each of the second locking feet **642** and the second buckling feet **643** includes a hook which is inserted into the corresponding locking hole **303** of the circuit board **300** and fixed to the circuit board **300**.
[0061] By providing the first buckling feet **633** and the second buckling feet **643**, the holding force of the metal cage **6** and the circuit board **300** can be improved, thereby reducing the risk of an external force applied to the cables pulling the metal cage **6** and separating the metal cage **6** from the circuit board **300**.
[0062] The above embodiments are only used to illustrate the present disclosure and not to limit the technical solutions described in the present disclosure. The understanding of this specification should be based on those skilled in the art. Descriptions of directions, such as “front”, “back”, “left”, “right”, “top” and “bottom”, although they have been described in detail in the above-mentioned embodiments of the present disclosure, those skilled in the art should understand that modifications or equivalent substitutions can still be made to the application, and all technical solutions and improvements that do not depart from the spirit and scope of the application should be covered by the claims of the application.

Claims

1. An electrical connector adapted to be at least partially mounted to a circuit board, the electrical connector comprising: an insulating body, the insulating body defining a mating slot; a plurality of first conductive terminals, the plurality of first conductive terminals comprising a plurality of first signal terminals and a plurality of first ground terminals, each of the first conductive terminals comprising a first mating portion extending into the mating slot, the first mating portions of the plurality of first conductive terminals being arranged in a first row; a plurality of second conductive terminals, the plurality of second conductive terminals comprising a plurality of second signal terminals and a plurality of second ground terminals, each of the second conductive terminals comprising a second mating portion extending into the mating slot, the second mating portions of the plurality of second conductive terminals being arranged in a second row; and a plurality of cables, the plurality of cables comprising a plurality of first cables electrically connected with the plurality of first signal terminals and a plurality of second cables electrically connected with the plurality of second signal terminals; wherein the plurality of first ground terminals and the plurality of second ground terminals are connected to each other, and are adapted to be electrically connected to the circuit board; wherein the plurality of first conductive terminals further comprise a plurality of first intermediate terminals, each of the first intermediate terminals comprises a first mounting foot for being mounted to the circuit board; wherein the plurality of second conductive terminals further comprise a plurality of second intermediate terminals, each of the second intermediate terminals comprises a second mounting foot for being mounted to the circuit board; wherein in a width direction of the insulating body, the plurality of first intermediate terminals are aligned with the plurality of first ground terminals, and the plurality of second intermediate terminals are aligned with the plurality of second ground terminals; wherein the electrical connector further comprises a first tail portion; the first tail portion is connected to at least one of the plurality of first ground terminals; and wherein the electrical connector further comprises a second tail portion; the second tail portion is connected to at least one of the plurality of second ground terminals.
2. The electrical connector according to claim 1, wherein the plurality of first signal terminals are divided into a plurality of first signal terminal groups, each of the first signal terminal groups comprises two of the first signal terminals, each first signal terminal group is located between two first ground terminals; the plurality of second signal terminals are divided into a plurality of second signal terminal groups, each of the second signal terminal groups comprises two of the second

signal terminals, each second signal terminal group is located between two second ground terminals; in the width direction of the insulating body, the plurality of first intermediate terminals are located between two groups of the plurality of first ground terminals, and the plurality of second intermediate terminals are located between two groups of the plurality of second ground terminals; the first tail portion is connected to at least one group of the two groups of the plurality of first ground terminals; and the second tail portion is connected to at least one group of the two groups of the plurality of second ground terminals.
3. The electrical connector according to claim 2, wherein the first tail portion is integrally connected with the at least one group of the two groups of the plurality of first ground terminals; and wherein the second tail portion is integrally connected with the at least one group of the two groups of the plurality of second ground terminals; wherein the first tail portion is in contact with the second tail portion.
4. The electrical connector according to claim 3, further comprising a ground mounting portion integrally extending from the first tail portion and/or the second tail portion; wherein the ground mounting portion is configured to be mounted to the circuit board.
5. The electrical connector according to claim 4, wherein the ground mounting portion comprises a through hole so that the ground mounting portion has a capability of being deformed so as to be engaged with a conductive hole of the circuit board.
6. The electrical connector according to claim 4, wherein the first signal terminals and the first cables are fixed by soldering or welding; wherein the second signal terminals and the second cables are fixed by soldering or welding.
7. The electrical connector according to claim 6, further comprising a first ground piece in contact with the plurality of first ground terminals so as to connect the plurality of first ground terminals in series, and a second ground piece in contact with the plurality of second ground terminals so as to connect the plurality of second ground terminals in series.
8. The electrical connector according to claim 6, wherein two first tail portions and two second tail portions are provided; the two groups of the plurality of first ground terminals are respectively integrally connected to the two first tail portions; the two groups of the plurality of second ground terminals are respectively integrally connected to the two second tail portions; and the two first tail portions are in contact with the two second tail portions, respectively.
9. The electrical connector according to claim 8, wherein the ground mounting portion comprises a first ground foot and a second ground foot, the first ground foot is integrally connected to one of the first tail portions, and the second ground foot is integrally connected to one of the second tail portions; wherein one group of the plurality of first ground terminals and one group of the plurality of second ground terminals are adapted to be electrically connected to the circuit board via the first ground foot, and an another group of the plurality of first ground terminals and an another group of the plurality of second ground terminals are adapted to be electrically connected to the circuit board via the second ground foot.
10. The electrical connector according to claim 9, wherein each of the first intermediate terminals comprises a first extension portion extending beyond the insulating body and a first bending portion bent from the first extension portion; the first bending portion comprises a first body portion, a first deflection portion bent from the first body portion, and a first end extending from the first deflection portion; the first end is parallel to the first body portion, and the first mounting foot is provided at the first end; wherein each of the second intermediate terminals comprises a second extension portion extending beyond the insulating body, and a second bending portion bent from the second extension portion; the second bending portion comprises a second body portion, a second deflection portion bent from the second body portion, and a second end extending from the second deflection portion; the second end is parallel to the second body portion, and the second mounting foot is provided at the second end; and wherein deflection directions of the first deflection portions of any two adjacent first intermediate terminals are opposite, the first mounting

foot and the first ground foot of any two adjacent first intermediate terminals are alternately arranged in two rows, deflection directions of the second deflection portions of any two adjacent second intermediate terminals are opposite, and the second mounting foot and the second ground foot of any two adjacent second intermediate terminals are alternately arranged in two rows.

11. The electrical connector according to claim 1, wherein the at least one of the plurality of first ground terminals is disposed in a first plane; the first tail portion is extended from the at least one of the plurality of first ground terminals to be disposed in a second plane, and the second plane is different from the first plane; and wherein the at least one of the plurality of second ground terminals is disposed in a third plane; the second tail portion is extended from the at least one of the plurality of second ground terminals to be disposed in a fourth plane, and the fourth plane is different from the third plane.

12. An electrical connector assembly, comprising: a first electrical connector; a second electrical connector; and a circuit board, the circuit board comprising a first side and a second side opposite to the first side; wherein the first electrical connector is mounted to the first side of the circuit board, and the second electrical connector is mounted to the second side of the circuit board; wherein the first electrical connector comprises: an insulating body, the insulating body defining a mating slot; a plurality of first conductive terminals, the plurality of first conductive terminals comprising a plurality of first signal terminals and a plurality of first ground terminals, each of the first conductive terminals comprising a first mating portion extending into the mating slot, the first mating portions of the plurality of first conductive terminals being arranged in a first row; a plurality of second conductive terminals, the plurality of second conductive terminals comprising a plurality of second signal terminals and a plurality of second ground terminals, each of the second conductive terminals comprising a second mating portion extending into the mating slot, the second mating portions of the plurality of second conductive terminals being arranged in a second row; and a plurality of cables, the plurality of cables comprising a plurality of first cables electrically connected with the plurality of first signal terminals and a plurality of second cables electrically connected with the plurality of second signal terminals; wherein the plurality of first ground terminals and the plurality of second ground terminals are connected to each other and are electrically connected to the circuit board; wherein the first electrical connector further comprises a first tail portion integrally connected with at least two of the plurality of first ground terminals and a second tail portion integrally connected with at least two of the plurality of second ground terminals; wherein the first tail portion is in contact with the second tail portion; and the first electrical connector further comprises a ground mounting portion integrally extending from one of the first tail portion and the second tail portion, and the ground mounting portion is mounted to the circuit board; a remaining one of the first tail portion and the second tail portion does not provide any ground mounting portion mounted to the circuit board; and the remaining one of the first tail portion and the second tail portion is electrically connected to the circuit board via the ground mounting portion provided on the one of the first tail portion and the second tail portion.

13. The electrical connector assembly according to claim 12, wherein the ground mounting portion comprises a through hole so that the ground mounting portion has a capability of being deformed so as to be engaged with a conductive hole of the circuit board.

14. The electrical connector assembly according to claim 12, wherein the plurality of first signal terminals are divided into a plurality of first signal terminal groups, each of the first signal terminal groups comprises two of the first signal terminals, each first signal terminal group is located between two first ground terminals, the first signal terminals and the first cables are fixed by soldering or welding; wherein the plurality of second signal terminals are divided into a plurality of second signal terminal groups, each of the second signal terminal groups comprises two of the second signal terminals, each second signal terminal group is located between two second ground terminals, the second signal terminals and the second cables are fixed by soldering or welding; and wherein each of the first tail portion and the second tail portion extends along a width direction of the insulating body, the plurality of first ground terminals are integrally connected to the first tail portion at intervals, and the plurality of second ground terminals are integrally connected to the second tail portion at intervals.

15. The electrical connector assembly according to claim 14, wherein the first electrical connector further comprises a first ground piece in contact with the plurality of first ground terminals so as to connect the plurality of first ground terminals in series, and a second ground piece in contact with the plurality of second ground terminals so as to connect the plurality of second ground terminals in series.

16. The electrical connector assembly according to claim 14, wherein the plurality of first conductive terminals further comprise a plurality of first intermediate terminals, each of the first intermediate terminals comprises a first mounting foot mounted to the circuit board; wherein the plurality of second conductive terminals further comprise a plurality of second intermediate terminals, each of the second intermediate terminals comprises a second mounting foot mounted to the circuit board; and wherein in the width direction, the plurality of first intermediate terminals are located between two groups of the plurality of first ground terminals, and the plurality of second intermediate terminals are located between two groups of the plurality of second ground terminals; the two groups of the plurality of first ground terminals are respectively integrally connected to two first tail portions, the two groups of the plurality of second ground terminals are respectively integrally connected to two second tail portions, and the two first tail portions are in contact with the two second tail portions, respectively.

17. The electrical connector assembly according to claim 16, wherein the ground mounting portion comprises a first ground foot and a second ground foot, the first ground foot is integrally connected to one of the first tail portions, and the second ground foot is integrally connected to one of the second tail portions; wherein one group of the plurality of first ground terminals and one group of the plurality of second ground terminals are electrically connected to the circuit board via the first ground foot, and an another group of the plurality of first ground terminals and an another group of the plurality of second ground terminals are electrically connected to the circuit board via the second ground foot.

18. The electrical connector assembly according to claim 17, wherein each of the first intermediate terminals comprises a first extension portion extending beyond the insulating body and a first bending portion bent from the first extension portion; the first bending portion comprises a first body portion, a first deflection portion bent from the first body portion, and a first end extending from the first deflection portion; the first end is parallel to the first body portion, and the first mounting foot is provided at the first end; wherein each of the second intermediate terminals comprises a second extension portion extending beyond the insulating body, and a second bending portion bent from the second extension portion; the second bending portion comprises a second body portion, a second deflection portion bent from the second body portion, and a second end extending from the second deflection portion; the second end is parallel to the second body portion, and the second mounting foot is provided at the second end; and wherein deflection directions of the first deflection portions of any two adjacent first intermediate terminals are opposite, the first mounting foot and the first ground foot of any two adjacent first intermediate terminals are alternately arranged in two rows, deflection directions of the second deflection portions of any two adjacent second intermediate terminals are opposite, and the second mounting foot and the second ground foot of any two adjacent second intermediate terminals are alternately arranged in two rows.

19. The electrical connector assembly according to claim 18, wherein the first electrical connector further comprises a first fixing block fixed to the plurality of first bending portions and the first ground foot, and a second fixing block fixed to the plurality of second bending portions and the second ground foot.

20. An electrical connector configured to be at least partially mounted to a circuit board, the electrical connector comprising: an insulating body, the insulating body defining a mating slot; a plurality of first conductive terminals, the plurality of first conductive terminals comprising a plurality of first signal terminals and a plurality of first ground terminals, each of the first conductive terminals comprising a first mating portion extending into the mating slot, the first mating portions of the plurality of first conductive terminals being arranged in a first row; a plurality of second conductive terminals, the plurality of second conductive terminals comprising a plurality of second signal terminals and a plurality of second ground terminals, each of the second conductive terminals comprising a second mating portion extending into the mating slot, the second mating portions of the plurality of second conductive terminals being arranged in a second row; a plurality of cables, the plurality of cables comprising a plurality of first cables connected to the plurality of first signal terminals and a plurality of second cables connected to the plurality of second signal terminals; a first tail portion connecting at least two of the plurality of first ground terminals; a second tail portion connecting at least two of the plurality of second ground terminals, the first tail portion being in contact with the second tail portion; and a ground mounting portion extending from one of the first tail portion and the second tail portion; wherein the ground mounting portion is configured to be mounted to the circuit board; a remaining one of the first tail portion and the second tail portion does not provide any ground mounting portion configured to be mounted to the circuit board; and the remaining one of the first tail portion and the second tail portion is electrically connected to the circuit board via the ground mounting portion provided on the one of the first tail portion and the second tail portion.